

Digital Product Passport with Product Carbon Footprint Calculation for the Wiring Harness

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Abstract—This work researches the current state of standardized methods for creation of the Digital Product Passport (DPP) and calculation of the Product Carbon Footprint (PCF) for automotive wiring harnesses. Focus is on defining a template with standardized data models for materials, processes and emissions to enable transparent, automatable, and cross-company exchangeable CO_2 balances. The aim is to enable the industry to fulfill forthcoming legislation for DPP creation with automated PCF calculation, provide guidance in PCF calculation method selection, proposing a generic data template for both Asset Administrations Shell (AAS) and Catena-X ecosystems as well as highlighting the challenges for the producing industry regarding data requirements and IT infrastructure.

Index Terms—AAS, Catena-X, DPP, PCF, KBL, VEC

I. INTRODUCTION

In December 2025, the EU Council and Parliament reached a provisional agreement on the new End-of-Life Vehicles (ELV) Regulation, marking a decisive step toward a circular automotive economy. The regulation mandates increasing shares of recycled plastics in new vehicles: at least 15 % within 6 years after entry into force, rising to 25 % within 10 years, with 20 % of the recycled content required to come from closed-loop recycling (materials recovered from ELVs). [1]

In order to reach and measure these goals, a central innovation is the mandatory **Circularity Vehicle Passport**. This Digital Product Passport (DPP) provides detailed, up-to-date data on material composition, recycled content percentages, and exact dismantling instructions, thus also fundamentally impacting how wiring harness systems are constructed, documented, disassembled, and recycled.

Extended Producer Responsibility (EPR) now holds OEMs financially and organizationally accountable for the full vehicle lifecycle, including net costs of collection and treatment. For the wiring harness industry — characterized by complex plastic insulation and multi-material designs — this creates new challenges regarding the creation and management of digital product passports, demanding a dedicated IT infrastructure for this purpose.

The principles of *Life Cycle Assessment*, particularly in the context of product-level assessments are applied to quantify *Product Carbon Footprint* (PCF) as the total direct and indirect greenhouse gas emissions associated with a product's life cycle stages. Emissions are typically expressed in tons of equivalent CO_2 (t CO_2 -eq), which includes carbon dioxide and other greenhouse gasses.

To structure this investigation, the following research questions are addressed:

- **RQ1:** What data standards and models are relevant for creating a Digital Product Passport for automotive wiring harnesses? (Sections II and III)
- **RQ2:** Which methods and data models should be applied for PCF calculation in the wiring harness domain? (Sections III and IV)
- **RQ3:** How can existing standardized submodel templates be adapted to enable a generic, interoperable DPP with integrated PCF data for wiring harnesses? (Section IV)

The scope of this paper is limited to the DPP during the engineering and production phases. The handling of the wiring harness DPP during product operation, i.e., while the harness is installed in the vehicle, is out of scope.

II. RELATED WORK

This section surveys the existing standards, data models, and frameworks relevant to DPP creation and PCF calculation for automotive wiring harnesses, thereby establishing the foundation for answering **RQ1**. The research project *DigiTain* (Digitalization for Sustainability)¹ developed processes, methods, and models for the fully digital product development and certification of sustainable electric drive architectures. A central objective of DigiTain was the integration of ecological and economical sustainability criteria from the early development phase. This included the use of digital technologies such as the Asset Administration Shell (AAS), Catena-X data spaces, and

¹<https://arena2036.de/digitain/>

Digital Product Passports (DPP) to enable seamless lifecycle data exchange, transparent Product Carbon Footprint (PCF) calculation, and traceability along the automotive value chain. The concepts were validated using a technology carrier for electric drive components [2].

A. ISO 14067

ISO 14067:2018 [3] specifies principles, requirements, and guidelines for the quantification and reporting of the carbon footprint of products, generally known as Product Carbon Footprints (PCF). The standard is based on the life cycle assessment (LCA) framework defined in ISO 14040 and ISO 14044 and focuses exclusively on the impact category climate change, expressed in carbon dioxide equivalents (CO_2eq). A PCF according to ISO 14067 requires a goal and scope definition, a life cycle inventory (LCI) analysis, a life cycle impact assessment (LCIA) and an interpretation of the results. The standard is not specific for any product or sector and results are highly dependent on the defined system boundaries, modeling choices, and impact assessment methods. This limits comparability of PCF results reported from different companies.

B. GHG Protocol

The GHG Protocol represents an international framework for the quantification and management of greenhouse gas (GHG) emissions [4]². It was jointly developed by the World Resources Institute (WRI) and the World Business Council for Sustainable Development (WBCSD) to provide standardized methodologies that ensure transparency, consistency, and comparability of emissions data across organizations and products. The GHG Protocol provides multiple standards primarily focusing on Corporate Carbon Footprints (CCF). Within CCF, emissions are classified into Scope 1 (direct emissions), Scope 2 (indirect energy emissions) and Scope 3 (value chain emissions). Besides CCF standards, a Product Lifecycle standard is provided, which is closely related to ISO 14067. In 2025, ISO and GHG Protocol announced a partnership to develop unified global standards for GHG emissions accounting [5]. The PACT Network provides an actual specification of the GHG data exchange protocol³.

C. LCA Study on automotive wiring harnesses

The *Transformations-Hub Leitungssatz* (Wiring Harness Transformation Hub) published a study in April 2025 with the full title *LCA-Studie zur Untersuchung der Umweltauswirkungen von Kfz-Leitungssätzen konventioneller und elektrischer Fahrzeuge* (LCA Study on the Environmental Impacts of Wiring Harnesses in Conventional and Electric Vehicles) [6]. The study provides a comparative cradle-to-cradle LCA of partial wiring harnesses in conventional internal combustion engine vehicles versus battery-electric vehicles to quantify environmental impacts, identify hotspots, reveal data gaps, and derive levers for impact reduction (e.g. material optimization

and recycling). It supports sustainability in the automotive supply chain, ecodesign, closed-loop recycling and EU regulations (e.g. regarding critical raw materials).

D. DIN DKE SPEC 99100

DIN DKE SPEC 99100 [7] defines data attributes for the digital battery passport required by the EU Battery Regulation (Regulation (EU) 2023/1542). The standard is specifically tailored to batteries and cannot be applied directly to wiring harnesses. However, adaptation as a structural template for other product groups like electronics, vehicles and so on is promising. The structured approach with mandatory and recommended attributes, machine-readable format, QR-code linkage etc. can serve as a blueprint for a wiring harness passport by creating a modular extension rather.

E. Ontologies

Research-driven ontologies for circular economy (CEON)⁴ [8] and DPP (DPPO)⁵ [9] have been defined and published as open-source to provide a general framework for creating and sharing data in product lifecycle management.

F. Catena-X

The Catena-X automotive data ecosystem [10] provides a collaborative, sovereign data infrastructure for resilient supply chain management. It has significantly advanced PCF standardization in the automotive industry through its PCF Rulebook [11] and the associated semantic data model.

The “*Product (Carbon) Footprint*” aspect model (`io.catenax.pcf/9.0.0`) [12] is a SAMM-based⁶ ontology designed for interoperable, machine-readable exchange of PCF data across the supply chain. It supports primary-data-driven PCF sharing via Eclipse Dataspace Components (EDC), digital twins, and APIs, while aligning with the Catena-X PCF Rulebook. The rulebook defines calculation rules, data quality requirements, allocation methods, and a cradle-to-gate focus, incorporating standards such as ISO 14067 and the GHG Protocol.

Key features include PCF values in $kg CO_2eq$ per declared unit, product and company identifiers, traceability information, BOM references, and support for multi-tier aggregation. The model is part of the Catena-X Use Case PCF (CX-0136) and integrates with other aspect models, including those for Digital Product Passports.

G. IMDS

The *International Material Data System* (IMDS) is the globally established platform for collecting and exchanging material composition data in the automotive industry. It serves as the central database for documenting the material and substance composition of components, semi-finished products, and raw materials to ensure compliance with regulations such

²<https://ghgprotocol.org/>

³<https://docs.carbon-transparency.org/data-exchange-protocol/>

⁴<https://liusemweb.github.io/CEON/>

⁵<https://liusemweb.github.io/DPPO/>

⁶<https://catenax-ev.github.io/docs/standards/CX-0003-SAMMSemanticAspectMetaModel>

as ELV, REACH, and GADSL, while supporting recyclability assessments.

IMDS captures sustainability-related information, including:

- Detailed material composition and substance declarations,
- Recycled content (post-industrial and post-consumer),
- Plastic identifiers for sorting and recycling.

All materials must be classified according to VDA 231-106. The “Norms/Standards” field should preferably reference public standards (DIN, EN, ISO, ASTM, etc.). TABLE I summarizes the main fields of the IMDS material data model.

Field	Necessity	Semantics / Rules / Naming Conventions
Name	Mandatory	Must describe the material itself (not trade name). Prefer public standards (EN, ISO, etc.). English mandatory; other languages optional.
Trade Name	Mandatory when specified on drawings.	Manufacturer’s commercial name (e.g., “Teflon”).
Internal Mat.-No.	Optional	Supplier-internal identifier (for internal use only).
Std. Mat.-No.	Mandatory for metals > 5g	Standard material number/symbol from public norms. Only for classes 1–4.
Symbol	Mandatory for polymers Class 5.x	ISO-based symbol (e.g., PE-LD, PA6-GF30)
Classification	Mandatory	VDA 231-106 based (Annex I to Rec. 001).
SCIP Material Category	Optional	EU SCIP database category (REACH).
Additional Material Characteristics	Optional	Free-text when no standard category fits.
Norms / Standards	Mandatory (most cases > 5g or Class 5)	Public or published in-house standards
Supplier	Optional	Manufacturer of the material.
Remark	Optional	Free-text comments.
Contains Recycled?	Mandatory	Yes/No (per Rec. 025).
Recyclate Content	Mandatory if Yes	Inorganic/fossil-based recyclate content.
Bio-based Content	Mandatory for applicable classes	Primary + secondary bio-based breakdown.
Composition (Substances)	Mandatory	Substances + jokers (max 9). Must sum exactly to 100% in final cured state.

TABLE I: IMDS Material Data Model Fields

For the upcoming EU ELV Regulation, IMDS provides a strong foundation for proving recycled plastic quotas. However, it lacks support for closed-loop verification, dynamic lifecycle updates, and detailed dismantling instructions required for the mandatory Circularity Vehicle Passport.

Since the release of IMDS version 15.0, PCF data can be entered directly into material and component datasets according to Recommendation 027, which is aligned with the Catena-X PCF Rulebook. This enables consistent reporting of product- and transport-related carbon footprints in $kg\ CO_2eq$ while maintaining data sovereignty. IMDS is expected to remain the primary system for supplier material declarations, but must

be complemented by additional digital tools such as Catena-X and Digital Product Passports for full lifecycle transparency and future ESG/CSRD requirements.

H. KBL and VEC

KBL (*KabelBaumListe* (Harness Description List), VDA-4964) and VEC (Vehicle Electric Container, VDA-4968) are engineering data standards for automotive electrical systems. Both are XML-based and support extensions via custom properties. The VEC model is also published as OWL2 Ontology for semantic web applications. They are widely used in the engineering process to describe the design of a wiring harness and in addition detailed technical component data in the case of the VEC. From a PCF perspective, the most relevant information they provide is the complete BOM hierarchy of the wiring harness, which is often omitted in classic PDM systems due to the complexity of the product. To limited extent, they also provide the materials involved and their weights.

The **KBL** arranges parts in a hierarchical BOM (Harness, Modules, Assemblies, Components) to attach variant management methods. The material can be defined for each part via a tuple composed of a reference system and a key. The reference system (e.g. IMDS or some standard) defines the semantic of the key. The primary focus of the KBL is to provide a build-to-print definition of the wiring harness. Therefore, substance-level details are deliberately omitted and delegated to external reference systems (e.g. IMDS).

The **VEC** provides the same capabilities as the KBL. In addition, it enables referencing material definitions from different reference systems simultaneously. It also comes with a formally defined and web-referencable ontology⁷. Furthermore, it allows the specification of material compositions, including mass fractions. However, for the identification of specific materials and substances, the VEC does not define its own reference system, but instead relies on external standards.

I. Asset Administration Shell (AAS)

The *Asset Administration Shell* (AAS) is the central standardized framework of Industry 4.0 for semantic modeling and interoperable exchange of asset-related data across the entire lifecycle. As the core implementation of the digital twin, it provides a modular, machine-readable information model consisting of submodels that describe properties, capabilities, states, services, and lifecycle information of physical or logical assets. Its standardized interfaces and serialization formats enable seamless data integration and automated consumption across heterogeneous systems and organizational boundaries.

Several AAS Submodel Templates (SMTs) are particularly relevant for Digital Product Passports in the automotive domain:

The SMT series *IDTA-02035-x: Digital Battery Passport*⁸ defines data structures compliant with DIN DKE SPEC 99100

⁷<https://ecad-wiki.prostep.org/specifications/vec/v220/vec-2.2.0-ontology.ttl>

⁸<https://github.com/admin-shell-io/submodel-templates/tree/main/published/Digital%20Battery%20Passport>

and the EU Battery Regulation (EU) 2023/1542. It comprises seven parts, many of which can be adapted for a Wiring Harness DPP.

The SMT *IDTA-02023: Carbon Footprint* [13] provides a standardized, extensible model for exchanging asset-related carbon footprint data along value chains. It combines a generic high-level footprint structure with sector-specific calculation rules, ensuring compatibility with ISO 14067, ISO 14040/14044, the GHG Protocol, and Catena-X PCF conventions. A calculation wizard for this submodel has been developed within the BaSyx framework⁹.

The SMT *IDTA-02011: Hierarchical Structures enabling Bills of Material* [14] defines a standardized representation of complex multi-level asset hierarchies, making it particularly suitable for modeling the multi-level Bill of Materials (BoM) and distributed asset structures of wiring harnesses.

J. DIN 182xx

The recently published DIN EN 182xx series is a suite of specifications on the technical infrastructure for DPP's, thus data handling, security and access for, but not on the actual content of the digital product passport, which comes from product-specific delegated acts.

These specifications are coordinated under DIN standards committee *NA 043-02-06 GA* and work together as a complete technical framework:

- DIN EN 18216 — Protocols for data exchange (secure formats and protocols for moving DPP data).
- DIN EN 18219 — Unique identifiers (rules for product IDs, economic operator IDs, and facility IDs).
- DIN EN 18220 — Data carrier (requirements for QR codes, RFID, etc., that link to a DPP).
- DIN EN 18221 — Data storage, archiving, and data persistence (decentralised storage rules and long-term durability).
- DIN EN 18222 — APIs for lifecycle management and searchability.
- DIN EN 18223 — System interoperability (overarching rules so different DPP systems can talk to each other).
- DIN EN 18239 — Management of user rights, IT security, and business secrets (access control, data protection, and responsibility transfer).
- DIN EN 18246 — Data authentication, reliability, and integrity (for tamper-proof and trustworthy data).

The goal is a secure, interoperable, decentralized, machine-readable, and future-proof DPP.

III. ANALYSIS

Building on the standards and data models presented in Section II, this section examines the practical challenges of applying them in the wiring harness domain. It addresses **RQ1** by analyzing material data representation across existing systems and **RQ2** by investigating PCF calculation methods,

data exchange mechanisms, and the difficulties arising along the supply and production chains.

A. Fragmented Identity Management

The wiring harness supply chain is highly decentralized, with stakeholders maintaining separate part numbering schemes and data repositories. The absence of a continuous cross-organizational identity management system results in data silos and inconsistent, poorly machine-readable product models¹⁰.

B. Material Attributes in IMDS, KBL, VEC and AAS

KBL and VEC provide limited support for material attributes, while IMDS is commonly used for regulatory compliance (e.g., REACH, ELV) although not mandated by the standards. Neither KBL nor VEC natively stores substance-level information such as CAS numbers. IMDS integration in VEC can be achieved by setting *referenceSystem* to “https://public.mdsystem.com/” and using the *key* field for the IMDS MDS ID. For richer material data, VEC allows custom properties on *GeneralTechnicalPartSpecification*. An generalized AAS Submodel based on *IDTA-02035-6: Digital Battery Passport – Part 6: Material Composition* [15] can leverage IMDS as a bridge, using its explicit support for IMDS references in KBL and VEC. This approach enables semantic alignment via CAS numbers and material hierarchies while minimizing custom extensions. However, inconsistent application of standards (primarily VDA 231-106 and ISO 1043) in IMDS often leads to semantic identification issues and must be dealt with.

C. PCF Calculation

PCF calculation according to ISO 14067 [3] involves multiple methodological choices, including functional unit definition, system boundaries, allocation rules, and impact assessment methods. In addition, results are affected by data quality and the use of secondary datasets, making the process complex and requiring strong LCA expertise, especially for smaller suppliers.

To address this, the Catena-X PCF Rulebook introduces sector-specific calculation rules that reduce methodological freedom and improve comparability across automotive supply chains. Complementary mechanisms such as Primary Data Share (PDS), Data Quality Ratings (DQR), and Product Verification Share (PVS) enhance transparency and trust in reported results while maintaining a standardized calculation approach. The PVS is determined according to the PCF Verification Framework [16].

Figure 1 presents a simple PCF calculation example for the process of *crimping*, where a *wire* and a *terminal* provide material inputs for the crimping process. All inputs and outputs are based on a unit of 1 piece of crimped cable (known as declared unit). The PCF of each flow is calculated as follows:

$$PCF_i = \text{ActivityData}_i \times \text{EmissionFactor}_i$$

⁹<https://github.com/eclipse-basyx/basyx-aas-web-ui/tree/main/aas-web-ui/src/pages/modules/DPPDemo>

¹⁰<https://vws4ls.github.io/>

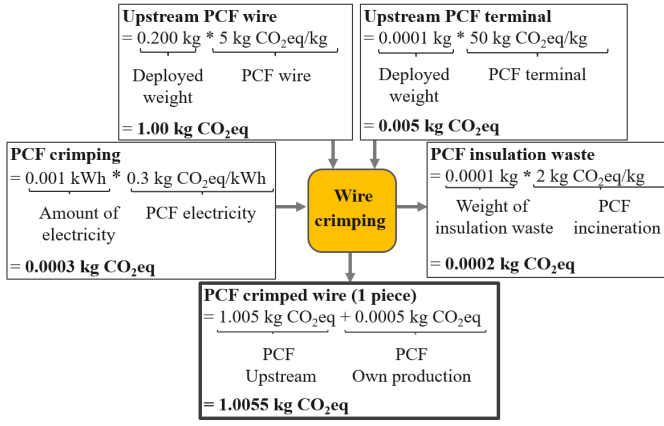


Fig. 1: PCF calculation example for wire crimping process

In the Catena-X ecosystem, the suppliers of the products are responsible for providing the respective PCF data to be used for the calculation. This example assumes that the suppliers are responsible in economic or operative terms for the outbound logistics. Emissions from transportation of the cable and terminal must be accounted by the suppliers in this case. The emission factor for the crimping process is derived from the electricity bill of the producer in this case. It must be noted that the latest version of the Catena-X PCF Rulebook requires infrastructure emissions from electricity generation to be included in the PCF calculation. Once all inputs have been accounted for, the emissions from outputs must be calculated. This example produces no direct emissions, but a material flow resulting from the cutting of cable insulation material. The treatment of this insulation waste results in GHG emissions, which must be allocated to the product system that generated the waste. In the example, the process of incinerating plastic waste without energy recovery is assumed and an emission factor from a secondary database is used.

This results in a total PCF of $1.0055 \text{ kg CO}_2\text{eq}$ for the produced crimped wire. The PCF of the wire crimping process including incineration of insulation waste amounts to $0.0005 \text{ kg CO}_2\text{eq}$, which is only 0.05% of the final result. The Catena-X PCF Rulebook allows for an exclusion of emissions if their sum represents less than 3% of the total PCF. [11] This cut-off rule may be applicable to a number of processes in the manufacturing of wiring harnesses, as the majority of emissions in a wire harness life cycle occur during material production according to the LCA study by the Transformations-Hub Leitungssatz [6].

A more detailed PCF calculation example according to the Catena-X PCF Rulebook including a screening analysis and application of the cut-off approach is available in [17]. Processing more comprehensive data (e.g. activity data and emission factors for a multitude of products) or complex calculations (e.g. for multi-output-processes involving allocation) may require dedicated PCF calculation software [18].

D. PCF Data Exchange

As described before, PCF calculations rely on activity data combined with emission factors, typically derived from IMDS material data supplemented by primary or database values. However, material descriptions in IMDS are often inconsistent or missing, requiring manual expert processing.

In Catena-X, PCF data is exchanged at component level. Each supplier provides a PCF dataset for its products in AAS format. Upstream PCF values from lower-tier suppliers are aggregated and added to the PCF of the supplier's own production processes.

This approach theoretically distributes the workload across the supply chain, with every tier responsible for its own products and thereby reducing gaps in LCI data and missing emission factors.

In practice, many Tier-1 and lower-tier suppliers still lack the expertise to generate reliable PCF data. A key task is therefore the systematic collection of standardized PCF datasets from component manufacturers. These must fulfill a defined minimum scope and be provided in JSON format according to the Catena-X PCF submodel (version 9.0.0).

TABLE II shows the relevant properties under *productLifeCycleStagesandEmissions* – *productionStage* for the wire crimping example. The *distributionStage* and *packagingStage* follow the same structure. All values are given in $\text{kg CO}_2\text{eq}$ per declared unit. Additional details and optional parameters are described in the official HTML documentation of the submodel [12]. However, these descriptions are often too technical for non-experts, highlighting the need to improve user-friendliness.

Property	Value	Necessity
pcfIncludingBiogenicUptake	1.0055	Mandatory
pcfExcludingBiogenicUptake	1.0055*	Mandatory
fossilGhgEmissions	1.0055*	Optional
biogenicNonCO2Emissions	0*	Optional
biogenicCO2Uptake	0*	Optional
landUseChangeGhgEmissions	0*	Optional
landManagementBiogenicCO2Emissions	0*	Optional
landManagementBiogenicCO2Removals	0*	Optional
aircraftGhgEmissions	0**	Optional

* No biogenic emissions in this example

** No aviation transport in this example

TABLE II: Catena-X PCF submodel properties under *productLifeCycleStagesandEmissions* – *productionStage* for the wire crimping example

Due to the high number of components and extreme variance in automotive wiring harnesses (often customer-specific), several thousand harnesses are produced daily in a single plant. This complexity demands fully automated data processing. Current software solutions are still limited, and a comprehensive, fully automated end-to-end solution is not yet available.

E. DPP during Engineering

In the engineering phase, relevant data must be provided in the engineering models and associated systems for the later creation of a DPP during the production process. In practice,

there are often gaps, rendering KBL and VEC engineering models largely unsuitable to be properly transferred to data space environments. The engineering tools and their capabilities are critical in this phase to ensure sufficient data quality and connection to data spaces like Catena-X.

F. DPP in the Supply Chain

Creating and continuously updating a DPP for wiring harnesses is highly challenging due to the complexity of global multi-tier supply chains and the high variability of product configurations. Key difficulties include the integration of heterogeneous data sources (ERP, PLM, supplier declarations, and material databases) that differ in structure, granularity, and accessibility. Semantic inconsistencies — especially in data from the IMDS — further complicate consistent interpretation, as identical materials and substances are often described using varying classification schemes and naming conventions. While standardization efforts such as those by the IEC offer promising solutions for semantic harmonization, practical adoption across the supply chain remains limited. Additionally, retrieving reliable supplier-specific data for PCF calculations is particularly difficult, as required information on materials, processes, and energy usage is frequently incomplete, proprietary, or unavailable at the necessary level of detail. Effective DPP implementation for wiring harnesses therefore demands robust data integration strategies, semantic alignment mechanisms, and incorporated solutions for handling data gaps and uncertainties.

G. DPP during the Production Process

Creating and continuously updating a Digital Product Passport (DPP) during production presents several challenges, especially for highly manual products such as wiring harnesses. Even though the continued modularisation of wire harnesses indicates that each sub harness will become smaller [19], hence easier to calculate a DPP for, the overall workload for the entire product will remain high.

The first challenge is real-time data acquisition on the shop floor. Wiring harness production involves numerous manual and semi-automated steps (cutting, crimping, assembly on form boards), requiring the capture of energy consumption, material usage, and scrap rates per process step and their attribution to individual product instances [20]. Dedicated identification and traceability mechanisms are essential to maintain a clear link between the physical product and its digital representation throughout the entire production sequence [21].

A second challenge is determining the appropriate data granularity. Whether the Product Carbon Footprint (PCF) is calculated at lot, order, or individual item level significantly influences the required measurement infrastructure and data volume. For many small and medium-sized enterprises in the wiring harness supply chain, item-level tracking remains impractical due to insufficiently digitalized shop floor systems [22]. This is exacerbated by fragmented and heteroge-

neous IT landscapes across ERP, PLM, MES, and quality management systems, often lacking a single source of truth [21].

The third challenge lies in the incremental enrichment of the DPP. Unlike a static document, the DPP must be progressively updated as the product moves through successive production stages. Especially if just regular, marginal, changes in the product as proposed by e.g. [23], make it necessary to repeatedly update the wire harness and its DPP. This demands robust update mechanisms capable of handling rework, quality deviations, and component substitutions while preserving data consistency. Although the Asset Administration Shell (AAS) provides a standardised framework via modular submodels [24], its practical integration with existing manufacturing execution systems remains a significant engineering challenge, particularly for companies with legacy infrastructure [20].

H. Catena-X data models

Catena-X provides established data models for both the Digital Product Passport (DPP) and Product Carbon Footprint (PCF). However, no dedicated model currently exists for representing a wiring harness as a system of components. Automotive wiring harnesses consist of numerous components with highly customized configurations per vehicle. Given that several thousand harnesses are produced daily in a single plant, fully automated data processing is essential. Current software solutions remain basic, with no clear path toward full automation.

The Catena-X PCF aspect model (version 9.0.0) is moderately to highly complex. While its detailed structure supports high-quality, traceable, and interoperable PCF data exchange in line with the PCF Rulebook V4, this granularity creates significant practical challenges. Properties such as *biogenicCarbonContent*, *biogenicNonCO2Emissions*, or *landUseChangeEmissions* often demand information that is impossible or extremely difficult to provide for many suppliers — especially lower-tier and non-European ones. As a result, full compliance risks leading to incomplete datasets, placeholder values, or the exclusion of smaller actors from the supply chain.

IV. SOLUTION

Based on the analysis in Section III, this section proposes concrete measures to address the identified challenges in DPP creation and PCF calculation for automotive wiring harnesses. It thereby completes the answer to **RQ2** through targeted adaptations of the Catena-X PCF data model and to **RQ3** by recommending AAS submodel adaptations, data quality mechanisms, and a pragmatic staged approach for production data collection.

A. Adaptation of the Catena-X PCF Aspect Model in the AAS

The Catena-X PCF Aspect Model should be mapped as a *SubmodelElementCollection* inside *ProductOrSectorSpecificRule* of the SMT IDTA-02023: *Carbon Footprint* [13].

The Catena-X PCF Aspect Model (v4) should be integrated as a *SubmodelElementCollection* within the *ProductOrSectorSpecificRule* element of the IDTA-02023 Carbon Footprint Submodel Template. Older Catena-X versions (v1–v3)

are considered deprecated. A new entry for *Catena-X v4* (`urn:samm:io.catenax.pcf:9.0.0`) must be added to the *PcfCalculationMethod* value list. Native Catena-X semanticIds are strongly preferred over ECLASS IRDIs to minimize external dependencies.

Value	ECLASS IRDI	Native <i>semanticId</i>
EN 15804	0173-1#07-ABU223	
GHG Protocol	0173-1#07-ABU221	
IEC TS 63058	0173-1#07-ABU222	
IEC 63366	0173-1#07-ACA792	
ISO 14040, 14044	0173-1#07-ABV505	
ISO 14067	0173-1#07-ABU218	
PEP Ecopassport	0173-1#07-ABU220	
PACT v1.0.1	0173-1#07-ACC004	
PACT v2.0.0	0173-1#07-ACC003	
PACT v3.0.0	0173-1#07-ACC012	
TFS v2	0173-1#07-ACC005	
TFS v3	0173-1#07-ACC010	
Catena-X v1	0173-1#07-ACC007	urn:samm:io.catenax.pcf:2.0.0
Catena-X v2	0173-1#07-ACC006	urn:samm:io.catenax.pcf:4.0.0
Catena-X v3	0173-1#07-ACC011	urn:samm:io.catenax.pcf:7.0.0
Catena-X v4		urn:samm:io.catenax.pcf:9.0.0
BS PAS 2050	0173-1#07-ACC008	
IEC 63372	0173-1#07-ACC019	

TABLE III: Value List for *PcfCalculationMethod*

Since no official HTTP namespace exists for Catena-X SAMM models, references shall use the `urn:samm:` scheme.

B. Adaptation of AAS Submodels

No neutral IDTA submodel for general material composition currently exists. The *IDTA-02035-6: Digital Battery Passport – Part 6: Material Composition* can serve as a suitable blueprint and can be generalized into a neutral *Product Material Composition* or *Asset Material Composition* submodel by applying the following replacements:

- *Battery* → *Product* or *Asset*
- *batteryChemistry* → *materialChemistry*
- *batteryMaterials* → *componentMaterials*
- Battery-specific components (cathode, anode, electrolyte) → generic or domain-specific terms (e.g., insulation, conductor, shielding, connector)
- *batteryMaterialLocation* → *componentLocation*

The resulting submodel enables direct mapping of KBL/VEC and IMDS material data while remaining fully compliant with existing AAS structures. It is recommended to publish this as a new IDTA-02xxxx series template.

C. Data Quality

Robust data quality and unambiguous semantic identification are essential for interoperability. While the rise of AI-driven multisensor quality inspection offers promising new opportunities to track quality parameters, the need for high quality data remains [25]. Engineering tools should implement direct IMDS import and proactive, automated validation of all semantic references and identifiers from the early stages of the product lifecycle. Before releasing an KBL or VEC engineering model, immediate user feedback and inline guidance should be provided for missing or invalid references. The VEC

ontology shall be used consistently as the primary reference framework. Where no web-resolvable identifiers exist, the following standards are recommended:

- Conductors: IEC 61360-4
- Insulation: ISO 1043 and ISO 6722

D. Handling Data Complexity

To lower the entry barrier for suppliers, the Catena-X PCF model should introduce clear fallback mechanisms, tiered data quality levels, guided validation, and sensible default values when primary data is unavailable.

E. Component Data Collection

Since many Tier-1 and lower-tier suppliers lack the expertise to generate and model PCF data in AAS, a user-friendly web-based interface within the Catena-X ecosystem is proposed. This front-end should allow OEMs to systematically request and collect PCF data directly from component manufacturers.

F. Production Data Collection

To capture energy consumption, material usage, and scrap rates at process level with acceptable effort, a hybrid approach is recommended: Process-level baseline values are defined during pre-series engineering. Selective real-time measurements are performed only on critical or high-impact process steps and used to continuously calibrate and improve the baseline data. This staged method balances accuracy requirements for DPP and PCF with practical implementation feasibility.

V. CONCLUSION

The analysis of the current state of the art for DPP creation in the wiring harness industry has shown that the primary barriers to effective implementation are *data quality* and *data complexity*. These challenges arise from fragmented data sources, inconsistent representations across the supply chain, and the high level of technical detail required for comprehensive DPPs. To overcome these barriers and enable scalable DPP adoption, we recommend the following key measures:

- 1) **Ensure dataspace interoperability** of wire harness artefacts by mandating the use of web-based, globally unique object identifiers within KBL and VEC descriptions. This can best be achieved by direct integration of engineering tools with the respective dataspace.
- 2) **Provide comprehensive tool support** for seamless integration of IMDS, automated data validation, and guided data collection starting from the earliest phases of the product lifecycle.
- 3) **Simplify Catena-X PCF calculation standard**, particularly with respect to highly demanding attributes (e.g. infrastructure emissions of electricity generation) that many suppliers cannot realistically provide under current conditions.
- 4) **Develop generalized SMTs** for the wiring harness DPP, building upon the proven structures of existing battery pass SMTs.

Within the framework of the *Robotik Challenge 2026*¹¹, an initial demonstrator prototype was developed on the Mnestix open-source framework¹². This prototype of a web-based DPP-Generator and -Viewer¹³ will be continuously extended and refined in subsequent project phases.

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¹¹<https://www.leitungssatz-hub.de/en/robotik-challenge/robotik-challenge-2026/begleitforschungs-und-digitalisierungsmodul/>

¹²<https://github.com/ARENA2036/THLS-DPP-mnestix-browser>

¹³<https://dpp-generator.arena2036.app>